A RESOLUTION COMPUTES THE COHOMOLOGY OF THE PRODUCT TOTAL COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- The product total complex of a double complex. Same chapter, the construction of the product total complex of a double complex, recalled in the proof. Supplies the total complex appearing in the statement and its differential.

Lemma 0.1. *Let $M^\bullet$ be a complex of abelian groups. Let*

$$\ldots \to A_2^\bullet \to A_1^\bullet \to A_0^\bullet \to M^\bullet \to 0$$

be an exact complex of complexes of abelian groups. Set $A^{p,q} = A_{-p}^q$ to obtain a double complex. Let $Tot_\pi(A^{\bullet,\bullet})$ be the product total complex associated to the double complex (see proof). Then the map $Tot_\pi(A^{\bullet,\bullet}) \to M^\bullet$ induced by $A_0^\bullet \to M^\bullet$ is a quasi-isomorphism.

Proof. Abbreviating $T^\bullet = \mathrm{Tot}_\pi(A^{\bullet,\bullet})$ we define

$$T^n = \prod\nolimits_{p+q=n} A^{p,q} = \prod\nolimits_{p+q=n} A_{-p}^q \quad \text{with} \quad \mathrm{d}_{T^\bullet}^n = \prod\nolimits_{n=p+q} (f_{-p}^q + (-1)^p d_{A_{-p}^\bullet}^q)$$

where $f_p^\bullet : A_p^\bullet \to A_{p-1}^\bullet$ are the maps of complexes in the lemma. We will show that $T^\bullet$ is acyclic when $M^\bullet$ is the zero complex. This will suffice by the following trick. Set $B_n^\bullet = A_{n+1}^\bullet$ and $B_0^\bullet = M^\bullet$. Then we have an exact sequence

$$\ldots \to B_2^\bullet \to B_1^\bullet \to B_0^\bullet \to 0 \to 0$$

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as in the lemma. Let $S^\bullet = \mathrm{Tot}_\pi(B^{\bullet,\bullet})$. Then there is an obvious short exact sequence of complexes

$$0 \to M^\bullet \to S^\bullet \to T^\bullet[1] \to 0$$

and we conclude by the long exact cohomology sequence. Some details omitted.

Assume $M^\bullet = 0$. We will show $H^0(T^\bullet) = 0$. The same argument works for other degrees. Let $x = (x_n) \in \mathrm{Ker}(d_{T^\bullet})$ map to $\xi \in H^0(T^\bullet)$ with $x_n \in A^{-n,n} = A^n_n$. Since $M^0 = 0$ we find that $x_0 = f^0_1(y_0)$ for some $y_0 \in A^0_1$. Then $x_1 - d^0_{A^\bullet_1}(y_0) = f^1_2(y_1)$ because it is mapped to zero by f^1_1 as x is a cocycle. for some $y_1 \in A^1_2$. Continuing, using induction, we find $y = (y_i) \in T^{-1}$ with $d_{T^\bullet}(y) = x$ as desired. $\square$